NxtPak AI — Packaging Optimization Report

Project: Tornado table blender | **Label:** PRELIMINARY GEOMETRIC PROTOTYPE | 2026-11-01 10:36 UTC

1. Product Specifications

Item	Value	Provenance / Label
Product Name	Tornado table blender	USER_ENTERED
Dimensions (L × W × H)	180.0 × 165.0 × 390.0 mm	USER_ENTERED
Packed Weight	3.2 kg	USER_ENTERED
Fragility Class	MEDIUM	Engineering Selection

2. Packaging Alternatives Comparison

Alternative	Outer Dims (mm)	Clearance	Vol. Util	Material	Rank
Tight Fit (Material-Optimized)	218.0 × 203.0 × 428.0	15.0 mm	67.35%	1795.54 cm³	ALTERNATIVE
Standard Fit (Balanced Protection)	258.0 × 243.0 × 468.0	25.0 mm	42.86%	7644.3 cm³	ALTERNATIVE
Protective Cushion (Fragility-Engineered)	312.0 × 297.0 × 522.0	40.0 mm	26.56%	18592.74 cm³	RECOMMENDED

3. ISO + ISTA Standards Evaluation Summary

Evaluated against: ISO registry v1 (16 entries) and ISTA registry v1 (14 entries), last verified 2026-11-01. No company-specific requirements loaded.

Standard	Check Type	Status	Evaluation Note
ISO 18602:2013	HUMAN_REVIEW	NEEDS_REVIEW	Sub-checks: volume-utilization=FAIL, material-savings=NOT_CONFIGURED. Overall conformance assessment requires human engineering review. A software PASS is never proof of ISO certification or successful physical testing. Physical-test standards require laboratory execution. No compliance certificates are generated.
ISO 18601:2013	HUMAN_REVIEW	NEEDS_REVIEW	ISO 18601:2013 requires qualified engineering review. Software cannot substitute for human assessment.
ISO 4180:2009	HUMAN_REVIEW	NEEDS_REVIEW	ISO 4180:2009 requires qualified engineering review. Software cannot substitute for human assessment.
ISO 2233:2000	PHYSICAL_TEST	NOT_VERIFIABLE	ISO 2233:2000 is a PHYSICAL_TEST standard. Software cannot execute or simulate physical tests. Status is always NOT_VERIFIABLE. A software PASS is never proof of ISO certification or successful physical testing. Physical-test standards require laboratory execution. No compliance certificates are generated.
ISO 2206:1987	PROGRAMMATIC	PASS	ISO 2206 face, edge, and corner naming convention is applied deterministically to all generated packaging geometry. Face 1=bottom, Face 2=top, Face 3=front, Face 4=back, Face 5=left, Face 6=right. [PROGRAMMATIC]

ISO 2248:1985	PHYSICAL_TEST	NOT_VERIFIABLE	ISO 2248:1985 is a PHYSICAL_TEST standard. Software cannot execute or simulate physical tests. Status is always NOT_VERIFIABLE. A software PASS is never proof of ISO certification or successful physical testing. Physical-test standards require laboratory execution. No compliance certificates are generated.
ISO 2244:2000	PHYSICAL_TEST	NOT_VERIFIABLE	ISO 2244:2000 is a PHYSICAL_TEST standard. Software cannot execute or simulate physical tests. Status is always NOT_VERIFIABLE. A software PASS is never proof of ISO certification or successful physical testing. Physical-test standards require laboratory execution. No compliance certificates are generated.
ISO 8768:1987	PHYSICAL_TEST	NOT_VERIFIABLE	ISO 8768:1987 is a PHYSICAL_TEST standard. Software cannot execute or simulate physical tests. Status is always NOT_VERIFIABLE. A software PASS is never proof of ISO certification or successful physical testing. Physical-test standards require laboratory execution. No compliance certificates are generated.

Suggested validation tests (to be performed in an accredited lab)

Summaries are for guidance. Use the current official ISTA procedure for testing.
No procedure suggested: enter product weight and shipping mode.

Mandatory Engineering Disclaimer:

A software PASS is never proof of ISO certification or of successful physical testing. All packaging geometry and savings estimates are preliminary engineering prototypes. Physical testing (drop, compression, vibration, environmental) in an accredited laboratory is strictly required before commercial production.